

Module A: Tabular Solution of the Simplex Method

Setup. For a standard-form LP

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0,$$

simplex moves from one **BFS** to another by changing the basis one column at a time.

Convert to Canonical Form After row operations that make the current basis matrix equal to the identity,

$$[I_m \quad Y \quad y_0] \implies x_B = y_0, x_N = 0.$$

Hence:

- if $y_0 \geq 0$, the current basic solution is a **BFS**;
- the current basis columns are exactly the columns that form I_m .

Reduced costs. If $A = [B \ D]$ and $c^\top = [c_B^\top \ c_N^\top]$, then

$$r_N^\top = \underbrace{c_N^\top}_{\text{cost increase by increasing } x_N} - \underbrace{c_B^\top B^{-1} D}_{\text{cost decrease by compensation}}.$$

In the tableau, the bottom row under the nonbasic columns is the reduced-cost row.

For a **minimization** problem:

- if all reduced costs are ≥ 0 , current BFS is optimal;
- if some $r_q < 0$, increasing x_q can decrease the objective.

Entering / leaving / pivot. Choose an entering variable x_q with $r_q < 0$ (often the most negative one).

Let the corresponding nonbasic column in the current tableau be $y_{\cdot q}$. To keep feasibility, use the ratio:

$$p = \arg \min_i \underbrace{\left\{ \frac{(y_0)_i}{y_{iq}} : y_{iq} > 0 \right\}}_{\text{MAX dist can go under } x \geq 0}.$$

Then:

- x_q enters the basis;
- the basic variable in row p leaves;
- pivot on (p, q) , i.e. use row operations so the new basis columns become I_m again.

Practice: Textbook 16.4 (tabular simplex)

Consider the linear program

$$\text{maximize } 2x_1 + x_2, \quad \text{subject to} \quad 0 \leq x_1 \leq 5, \quad 0 \leq x_2 \leq 7, \quad x_1 + x_2 \leq 9.$$

1. Convert the problem to standard form using slack variables.
2. Starting from the slack-variable basis, solve the LP by the **tableau simplex method**.
3. State the optimal solution and optimal objective value for the original LP.

Standard primal/dual pair (lecture form). For

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0,$$

the dual is

$$\max \lambda^\top b \quad \text{s.t.} \quad \lambda^\top A \leq c^\top.$$

Min–max intuition. The primal can be written as

$$\min_{x \geq 0} \max_{\lambda} \left(c^\top x - \lambda^\top (Ax - b) \right).$$

If $Ax \neq b$, the inner maximization can blow up, so any primal optimum must satisfy $Ax = b$.

Swapping the order gives

$$\max_{\lambda} \min_{x \geq 0} \left(c^\top x - \lambda^\top (Ax - b) \right).$$

For fixed λ , the inner minimum is finite only if

$$c^\top - \lambda^\top A \geq 0,$$

which yields the dual constraints.

Useful max– $\leq$ form. If the primal is

$$\max c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0,$$

then the dual is

$$\min b^\top \lambda \quad \text{s.t.} \quad A^\top \lambda \geq c, \quad \lambda \geq 0.$$

Weak duality. For any primal-feasible x and dual-feasible y ,

$$c^\top x \leq b^\top y$$

That is:

- a dual-feasible point gives an **upper bound** on a primal maximization problem;
- a primal-feasible point gives a **lower bound** on a dual minimization problem.

Strong duality / Duality Theorem If the primal problem has an optimal solution, then so does the dual, and the optimal values of their respective objective functions are equal.

Practice: Textbook 17.3 (duality)

Consider the linear program

$$\text{maximize } 2x_1 + 3x_2$$

subject to

$$x_1 + 2x_2 \leq 4, \quad 2x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0.$$

1. Solve the primal problem using tabular simplex.
2. Write down the dual problem.
3. Solve the dual and verify that the two optimal objective values coincide.